



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Collected 90 Falcon 9 launch records via the SpaceX REST API and Wikipedia web scraping, then wrangled them into a clean, labeled dataset (Class = 1 landed / 0 did not land)
- Performed EDA with visualization, SQL, Folium, and Plotly Dash to uncover patterns between launch site, orbit type, payload mass, and landing success
- Built and tuned four classification models (Logistic Regression, SVM, Decision Tree, KNN) via GridSearchCV to predict first-stage landing outcomes
- Key result: overall historical success rate is 66.7%; Decision Tree was the best model at 88.9% test accuracy; KSC LC-39A has the highest launch success rate (76.9%)

Introduction

- SpaceX advertises Falcon 9 launches at \$62M, versus \$165M+ from other providers — largely because SpaceX reuses the first stage instead of discarding it
- If we can predict whether a launch's first stage will land successfully, we can estimate the cost of a launch in advance
- This matters for any competitor bidding against SpaceX for a launch contract, and for understanding which factors (launch site, orbit, payload, booster version) drive landing success
- Central question: Can we build a model that reliably predicts Falcon 9 first-stage landing success from pre-launch data?



Section 1

Methodology

Methodology

- Data collection methodology:
 - SpaceX REST API (structured records) + Wikipedia web scraping (cross-check), merged into 90 Falcon 9 launches
- Perform data wrangling
 - Imputed missing payload mass; mapped raw landing outcomes to a binary Class label (1 = landed)
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - Logistic Regression, SVM, Decision Tree, KNN tuned via 10-fold GridSearchCV; compared by held-out test accuracy

Data Collection

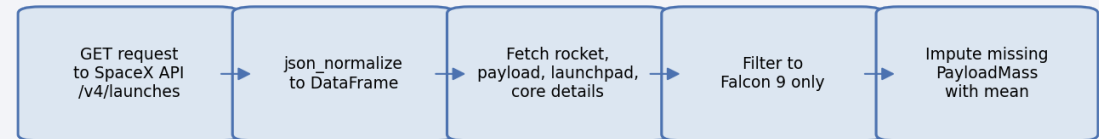
Two complementary data sources were combined: SpaceX's own REST API (structured launch records) and Wikipedia's Falcon 9 launch table (web scraping) for cross-validation

Both pipelines converge on a common schema: flight number, date, booster version, launch site, payload mass, orbit, and landing outcome

GitHub: <https://github.com/AbhinavJag/spacex-falcon9-capstone>

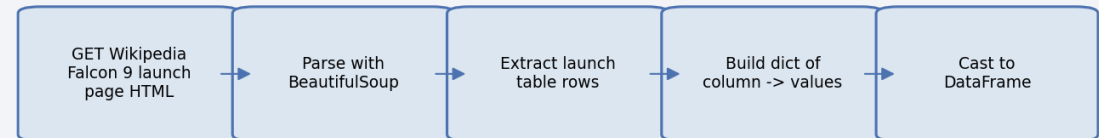
Data Collection – SpaceX API

- GET request to `api.spacexdata.com/v4/launches/past`, parsed with `pandas.json_normalize()`
- Follow-up API calls resolve rocket, payload, launchpad, and core IDs into readable fields
- Filtered to Falcon 9 only (dropped Falcon 1), yielding 90 launch records
- GitHub: <https://github.com/AbhinavJag/spacex-falcon9-capstone>



Data Collection - Scraping

- Requested the Wikipedia 'List of Falcon 9 and Falcon Heavy launches' page and parsed it with BeautifulSoup
- Extracted each launch table row into a dictionary, then cast to a DataFrame as a cross-check against the API dataset
- GitHub:
<https://github.com/AbhinavJag/spacex-falcon9->



Data Wrangling

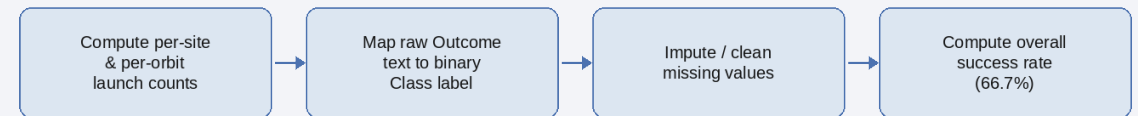
Computed launch counts per site (CCAFS SLC 40: 55, KSC LC-39A: 22, VAFB SLC 4E: 13) and per orbit (GTO: 27, ISS: 21, VLEO: 14, etc.)

Mapped the raw Outcome column (e.g. 'True ASDS', 'False Ocean') into a binary Class label: 1 = successful landing, 0 = not

Resulting success rate across all 90 launches: 66.7%

GitHub:

<https://github.com/AbhinavJag/spacex-falcon9-capstone>

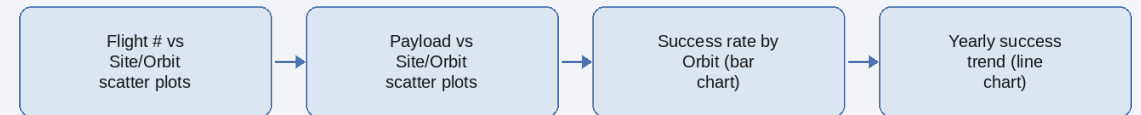


EDA with Data Visualization

Plotted Flight Number & Payload Mass against Launch Site and Orbit Type to look for spatial/orbital patterns in success

Plotted success rate by orbit type as a bar chart, and yearly success trend as a line chart, to see how SpaceX's landing capability improved over time

GitHub:
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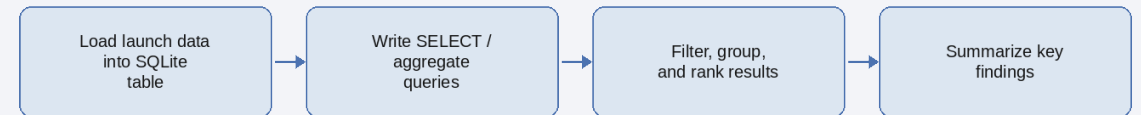
EDA with SQL

Queried unique launch sites, CCA-prefixed site records, and total/average payload mass by customer & booster version

Found the first successful ground-pad landing date (2015-12-22) and boosters that succeeded on drone-ship landings with mid-range payloads

Ranked landing outcome counts between 2010-06-04 and 2017-03-20 to see the mix of attempt types over SpaceX's early years

GitHub:
<https://github.com/AbhinavJag/space-x-falcon9-capstone>



Interactive Visual Analytics: Folium Map & Plotly Dash

Added a colored circle + label marker for each of the 4 launch sites, plus a MarkerCluster of every individual launch color-coded green (success) / red (failure)

Calculated real distances from CCAFS SLC-40 to its nearest coastline, railway, highway, and city, and drew PolyLines to each

Paired with a Plotly Dash app: a launch-site dropdown drives a success-count pie chart, and a payload-range slider drives a payload-vs-outcome scatter plot (see following slides for the dashboard walkthrough)

GitHub: <https://github.com/AbhinavJag/spacex-falcon9-capstone>

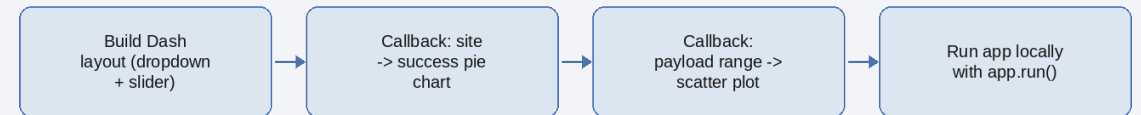
Build a Dashboard with Plotly Dash

Added a launch-site dropdown driving a pie chart of success counts, and a payload range slider driving a payload-vs-outcome scatter plot

These interactions let a user isolate any one site or payload band and immediately see its success pattern, colored by booster version

GitHub:

<https://github.com/AbhinavJag/spacex-falcon9-capstone>



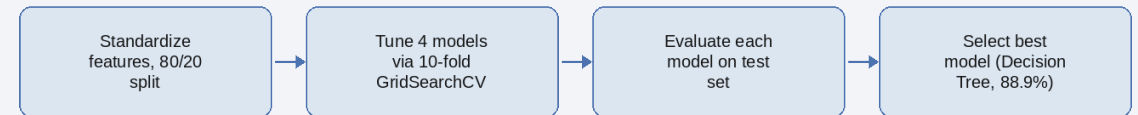
Predictive Analysis (Classification)

Standardized features, split into 80/20 train/test (18 test samples), then tuned Logistic Regression, SVM, Decision Tree, and KNN via 10-fold GridSearchCV

Best validation accuracy came from Decision Tree (88.9%); on the held-out test set, Decision Tree also led at 88.9%, versus 83.3% for the other three models

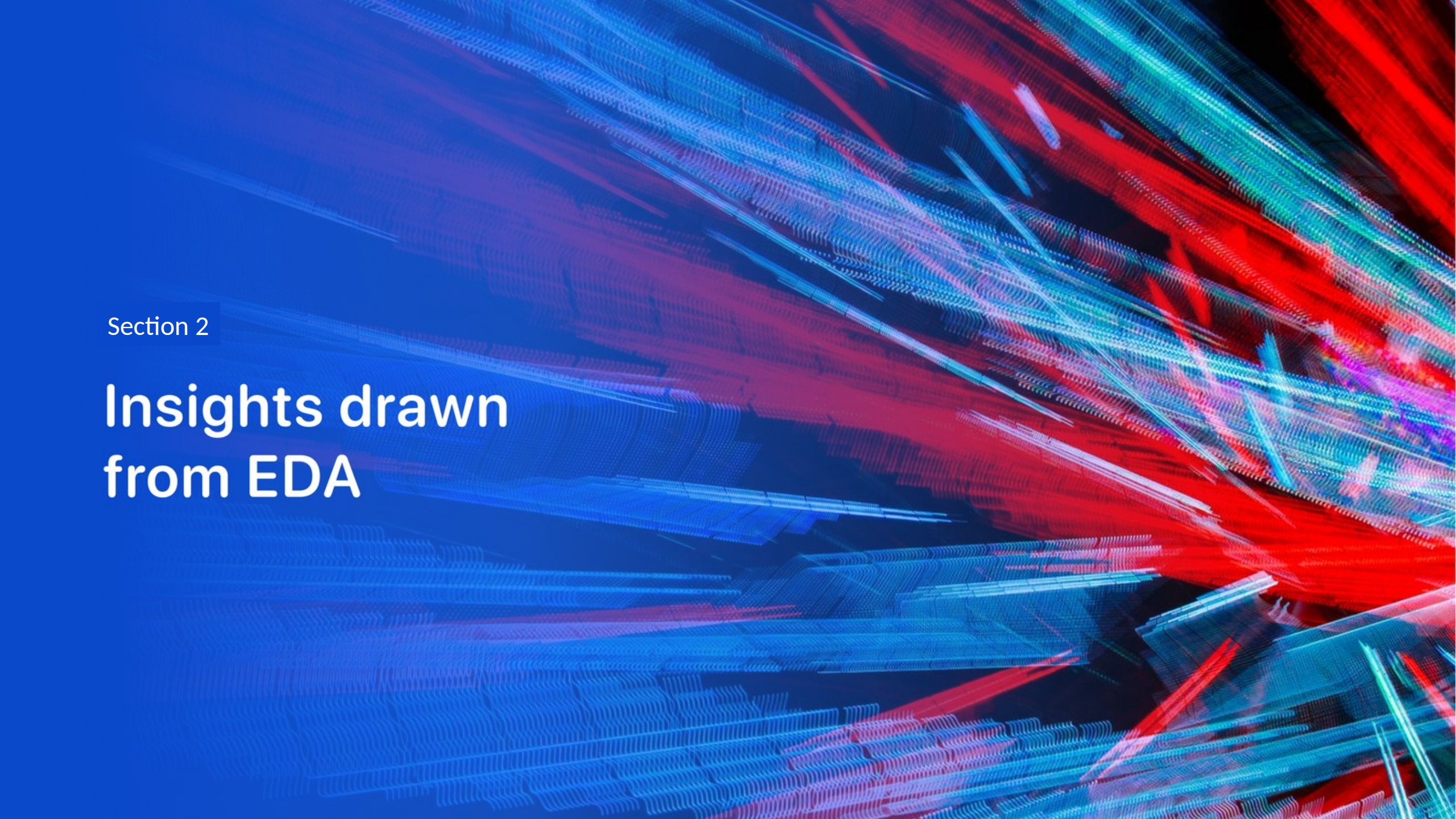
Takeaway: a competitor could use this model on published pre-launch specs to estimate SpaceX's reuse-driven cost advantage on a bid before a contract is signed, turning a guess into a ~89%-reliable, data-backed number

GitHub:
<https://github.com/AbhinavJag/spacex-falcon9-capstone>



Results

- EDA: KSC LC-39A has the best success rate (76.9%); GEO/HEO/SSO/ES-L1 orbits are 100% successful historically; success rate climbs steadily by launch year
- Interactive analytics: Folium confirms all launch sites sit near the coast at low latitude; Dash dashboard shows KSC LC-39A leading in both count and rate of successful launches
- Predictive analysis: Decision Tree is the best-performing model at 88.9% test accuracy

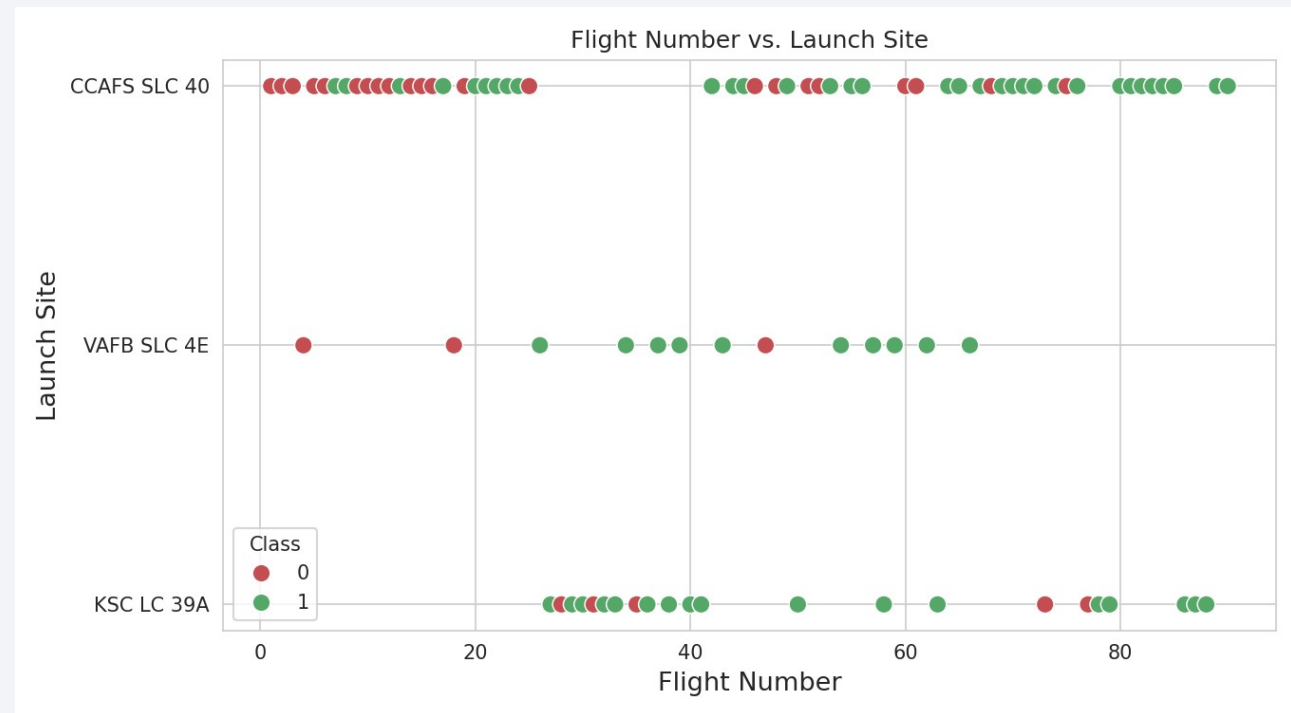


Section 2

Insights drawn from EDA

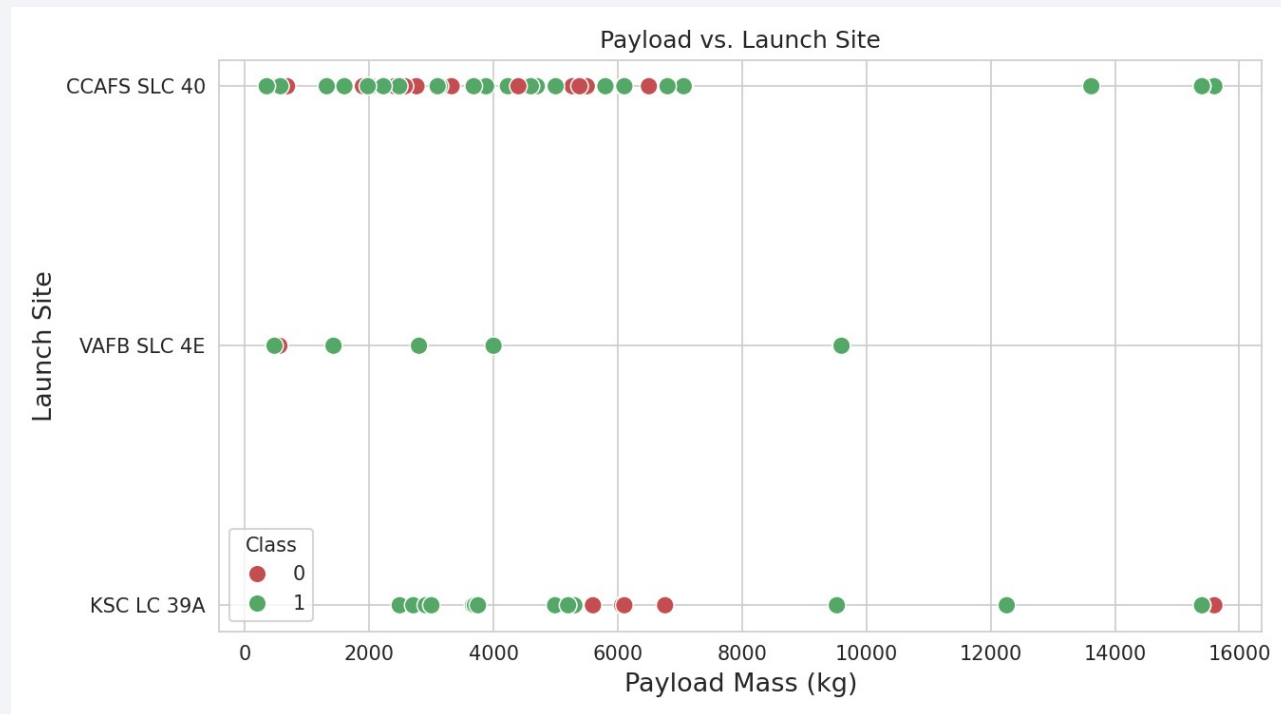
Flight Number vs. Launch Site

- Later flights (higher flight number) cluster toward successful landings (green/upper points)
- CCAFS SLC 40 has the most launches overall; KSC LC-39A shows a high concentration of successes



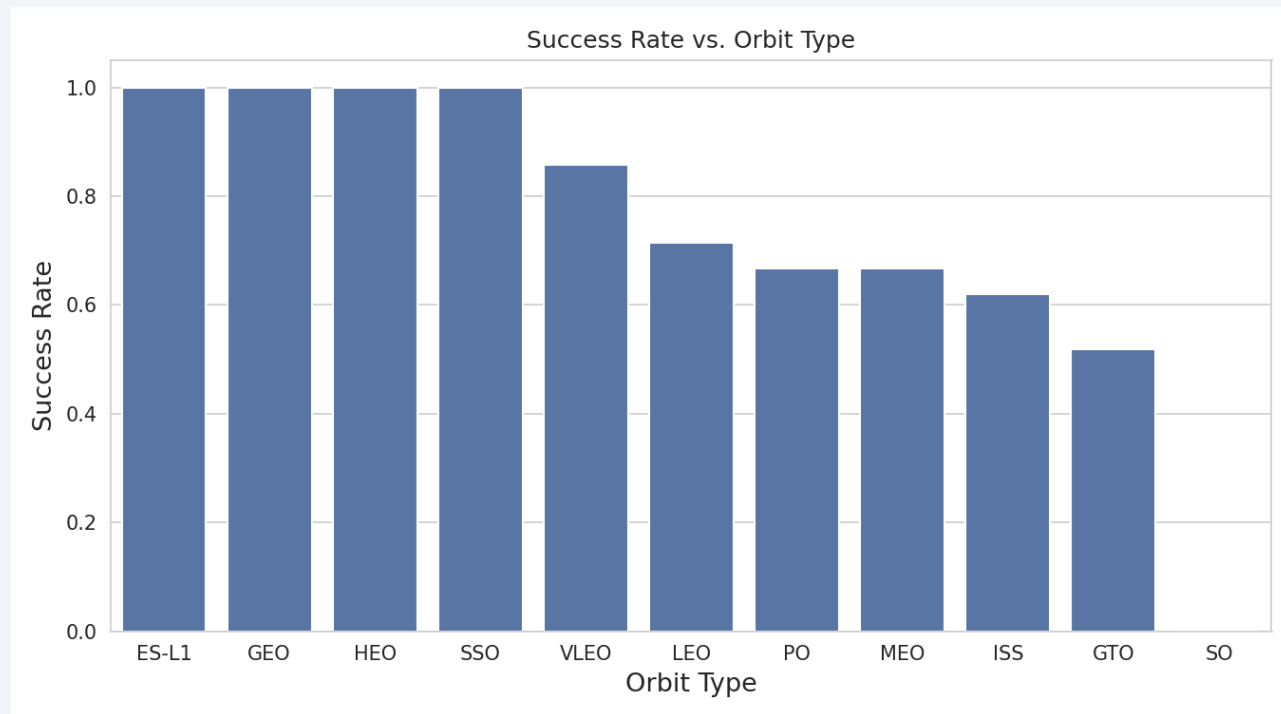
Payload vs. Launch Site

- VAFB SLC 4E only carries lighter payloads; CCAFS sites handle the full payload range
- Very heavy payloads (>8000 kg) skew toward launch failures across all sites



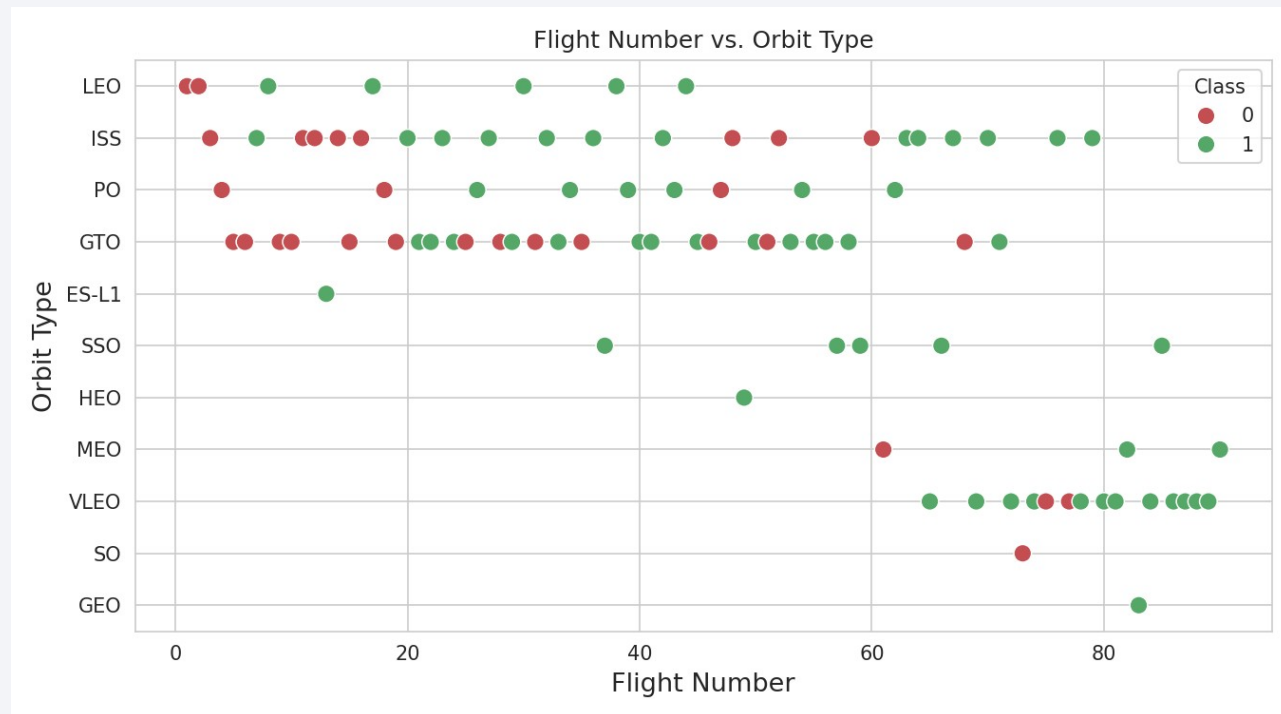
Success Rate vs. Orbit Type

- ES-L1, GEO, HEO, and SSO orbits have a 100% historical success rate (small sample sizes)
- GTO, the most common orbit, has the lowest success rate among the frequently-used orbit types (51.9%)



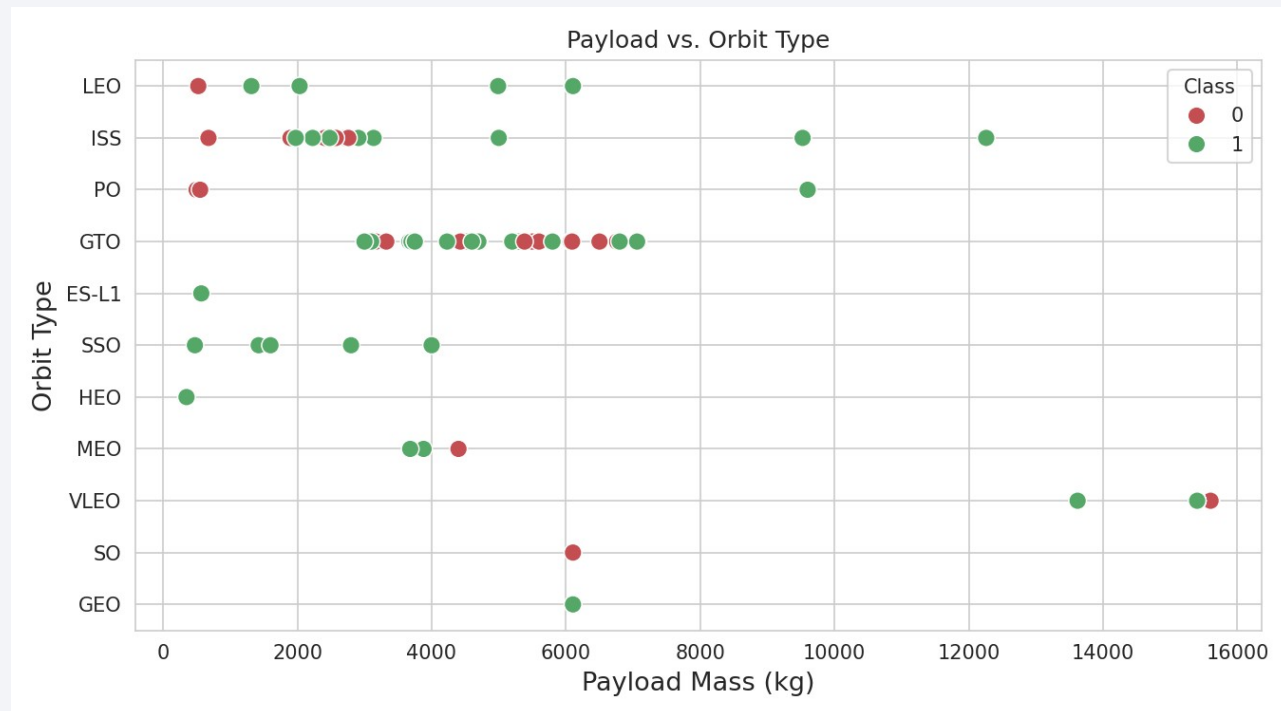
Flight Number vs. Orbit Type

- Early flights (low flight number) targeted mostly LEO/ISS/GTO; VLEO only appears in recent, higher-numbered flights
- Success (green) becomes far more common as flight number increases, regardless of orbit



Payload vs. Orbit Type

- GTO missions carry a wide payload range (2000-8000 kg) with mixed outcomes
- ISS and LEO payloads cluster at lower mass and skew successful



Launch Success Yearly Trend

- Success rate was near 0 in 2013-2015, climbed sharply through 2017-2018, and has stayed high (>80%) since 2019
- This trend reflects SpaceX's iterative booster design improvements (v1.0 -> v1.1 -> FT -> B4/B5)



All Launch Site Names

Query: `SELECT DISTINCT Launch_Site FROM SPACEXTABLE`

Result: CCAFS LC-40, CCAFS SLC-40, KSC LC-39A, VAFB SLC-4E

4 distinct launch sites recorded across the full 101-row raw launch history

Launch Site Names Begin with 'CCA'

Query: `SELECT * FROM SPACEXTABLE WHERE Launch_Site LIKE 'CCA%' LIMIT 5`

Returns the first 5 launches from Cape Canaveral (CCAFS LC-40), 2010-2013

All 5 were early Falcon 9 v1.0 missions for NASA COTS/CRS demo flights

Total Payload Mass

Query: `SELECT SUM(PAYLOAD_MASS__KG_) FROM SPACEXTABLE
WHERE Customer = 'NASA (CRS)'`

Result: 45,596 kg total payload mass carried for NASA's
Commercial Resupply Services contract

Average Payload Mass by F9 v1.1

Query: `SELECT AVG(PAYLOAD_MASS_KG_) FROM SPACEXTABLE
WHERE Booster_Version LIKE 'F9 v1.1%'`

Result: 2,534.67 kg average payload mass for the F9 v1.1 booster version

First Successful Ground Landing Date

Query: `MIN(Date) WHERE Landing_Outcome = 'Success (ground pad)'`

Result: 2015-12-22 — the first-ever successful Falcon 9 ground-pad landing (Orbcomm-2 mission)

Successful Drone Ship Landing with Payload between 4000 and 6000

Query: Booster_Version WHERE Landing_Outcome = 'Success (drone ship)' AND payload BETWEEN 4000-6000 kg

Result: F9 FT B1022, F9 FT B1026, F9 FT B1021.2, F9 FT B1031.2

4 boosters successfully landed on a drone ship carrying a mid-range payload

Total Number of Successful and Failure Mission Outcomes

Query: COUNT(*) grouped by Success/Failure landing outcome

Result: 61 successful landing outcomes, 40 failed landing outcomes (101 total)

Overall landing success rate across the full raw dataset: 60.4% (this raw 101-row SQL table includes Falcon Heavy and pre-cleaning duplicate records; the 66.7% figure used elsewhere reflects the cleaned, Falcon-9-only, 90-launch ML dataset)

Boosters Carried Maximum Payload

Query: `Booster_Version WHERE PAYLOAD_MASS = MAX(PAYLOAD_MASS)`

Result: 12 B5-generation boosters tied for the maximum payload mass carried (e.g. B1048.4/.5, B1049.4/.5/.7, B1051.3/.4/.6, B1056.4, B1058.3, B1060.2/.3)

2015 Launch Records

Query: Failed drone-ship landings, booster & site, WHERE year = 2015

Result: F9 v1.1 B1012 and F9 v1.1 B1015, both launched from CCAFS LC-40, both 'Failure (drone ship)'

These were among SpaceX's earliest drone-ship landing attempts, before the technique was perfected

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

Query: COUNT(*) per landing outcome between 2010-06-04 and 2017-03-20, ranked descending

No attempt: 10 | Success (drone ship): 5 | Failure (drone ship): 5

Success (ground pad): 3 | Controlled (ocean): 3 | Failure (parachute): 2

Uncontrolled (ocean): 2 | Precluded (drone ship): 1

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a dark blue sky with stars and a view of the Earth's surface from space. The Earth's surface is mostly dark, with a thin layer of atmosphere visible along the horizon. The city lights are concentrated in the lower right quadrant, showing a dense network of urban areas. The text "Section 3" is overlaid on the left side of the image.

Section 3

Launch Sites Proximities Analysis

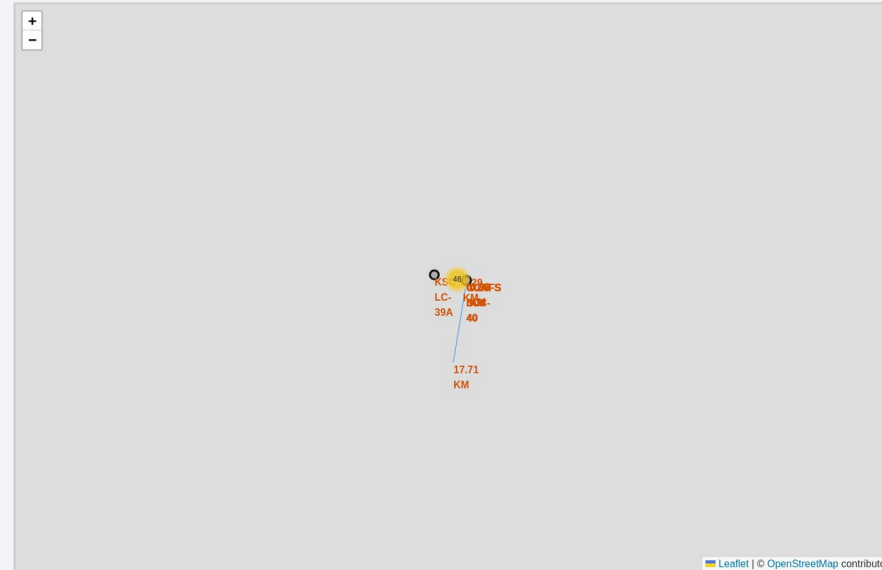
All Launch Sites on a Global Map



The interactive map (spacex_launch_site_map.html) marks all 4 launch sites with circles and labels. All sites sit near the coast at low latitude, consistent with rocket safety/range requirements.

GitHub: <https://github.com/AbhinavJag/spacex-falcon9-capstone>

Color-Labeled Launch Outcomes by Site



MarkerCluster groups launches per site, colored green (success) / red (failure). Zoomed on the Florida sites (CCAFS LC-40/SLC-40 and KSC LC-39A), individual launch outcomes are visible.

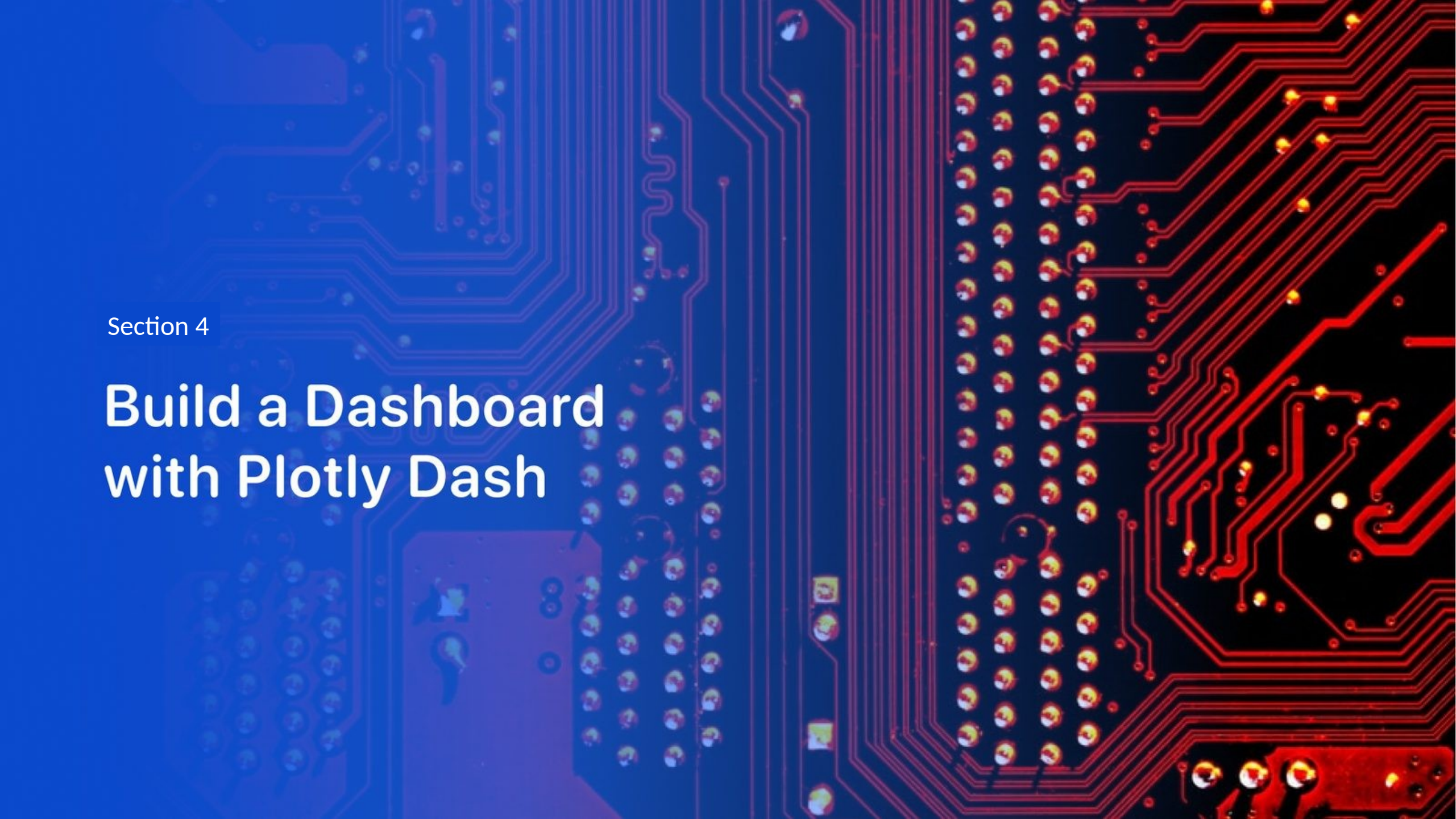
GitHub: <https://github.com/AbhinavJag/spacex-falcon9-capstone>

CCAFS SLC-40 Proximity Analysis



From CCAFS SLC-40: nearest coastline 0.89 km, nearest railway 1.29 km, nearest highway 0.60 km, nearest city (Cape Canaveral) 17.71 km — launch sites keep a buffer from populated areas but stay close to transport infrastructure.

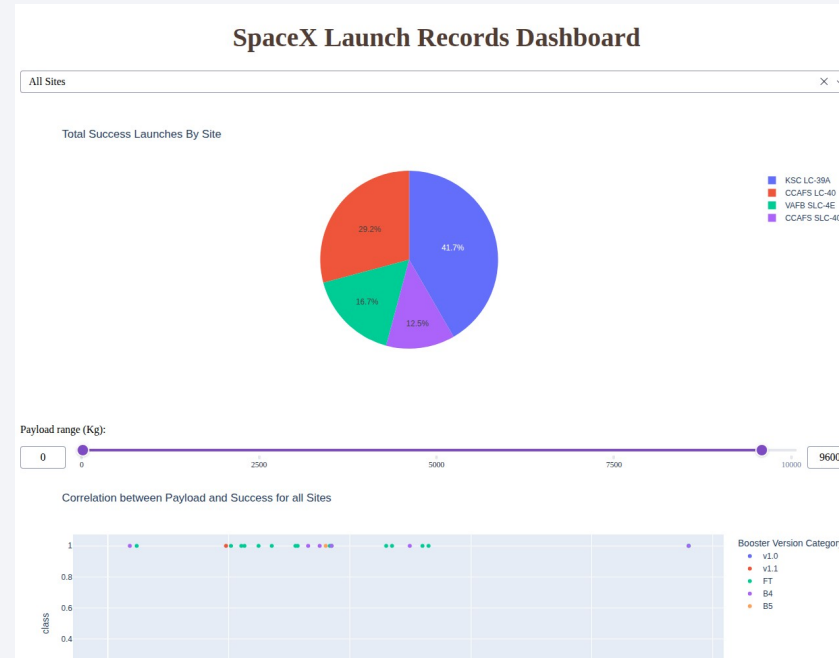
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Section 4

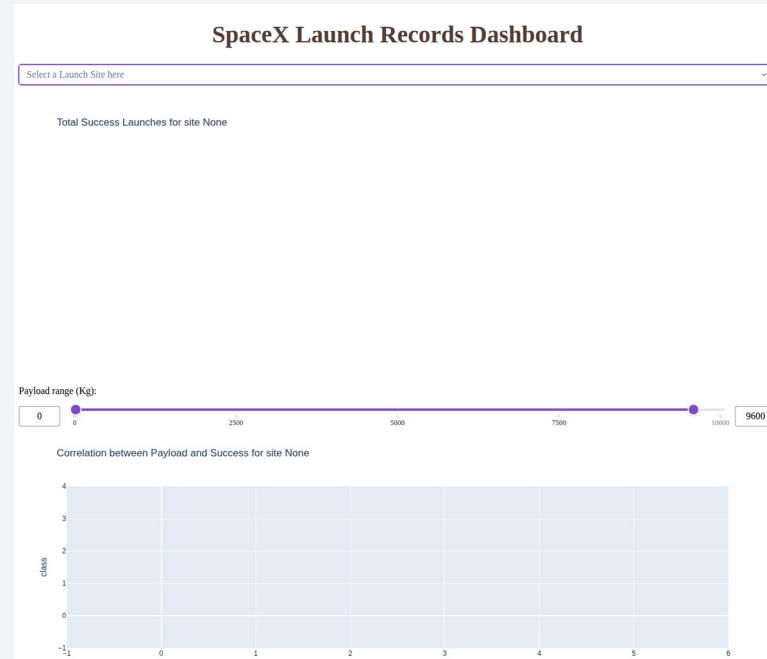
Build a Dashboard with Plotly Dash

Launch Success Count by Site (All Sites)



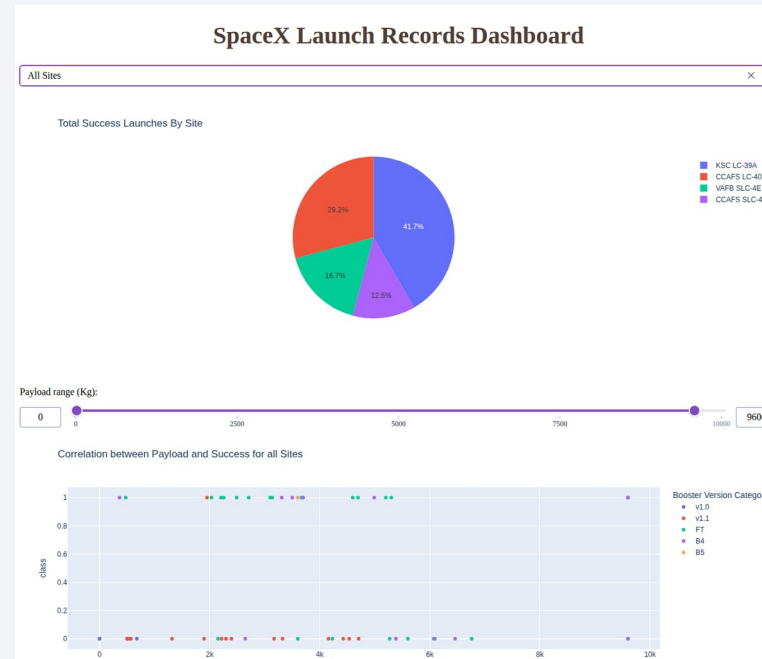
KSC LC-39A accounts for 41.7% of all successful launches — the largest share of any single site.

Launch Site with Highest Success Ratio (KSC LC-39A)



Selecting KSC LC-39A in the dropdown shows its individual success/failure split: 76.9% success rate, the best of the four sites.

Payload vs. Launch Outcome, Colored by Booster Version

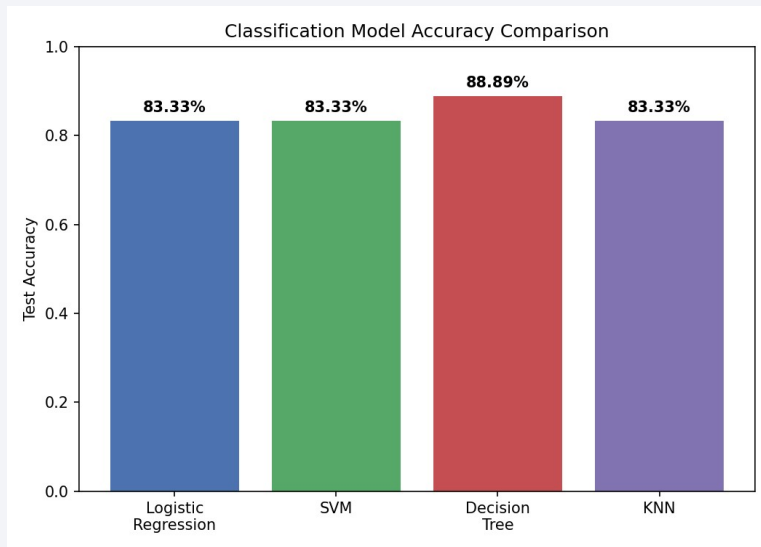


Across the full payload range, newer booster versions (FT, B4, B5 — green/purple/orange) cluster toward successful landings (class=1), while v1.0/v1.1 (blue/red) cluster toward failures (class=0).

Section 5

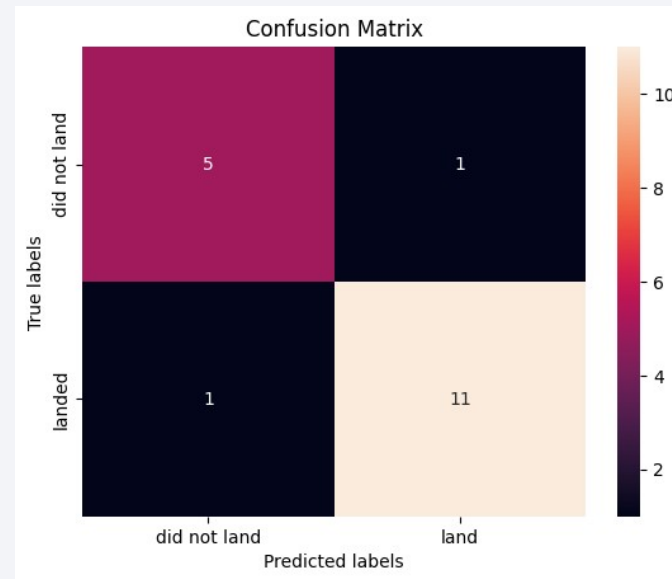
Predictive Analysis (Classification)

Classification Accuracy



Decision Tree achieved the highest test accuracy (88.9%); Logistic Regression, SVM, and KNN tied at 83.3%.

Confusion Matrix



Decision Tree confusion matrix on the 18-sample test set: 5 true negatives, 11 true positives, 1 false positive, 1 false negative.

Conclusions

Falcon 9 first-stage landing success has improved dramatically over time, from near 0% in 2013-2015 to consistently above 80% since 2019, as booster versions matured (v1.0 -> FT -> B5)

KSC LC-39A is the strongest-performing launch site (76.9% success rate); GTO missions, while the most common, have a comparatively lower success rate (51.9%) than LEO/ISS/VLEO

A Decision Tree classifier predicts landing success with 88.9% test accuracy, outperforming Logistic Regression, SVM, and KNN (all 83.3%) on this dataset

Payload mass and booster version are strong predictors: newer boosters (FT, B4, B5) succeed far more often than early v1.0/v1.1 versions across all payload ranges

These findings support using booster generation, launch site, and orbit type as leading indicators when estimating launch cost and risk for competitive bidding

Appendix

Full completed notebooks: data collection (API + web scraping), data wrangling, EDA (visualization + SQL), Folium interactive map, Plotly Dash dashboard, and ML prediction pipeline

Key verified figures: 90 Falcon 9 launches analyzed, 66.7% overall success rate, 18-sample test set, Decision Tree best model (88.9% test accuracy)

GitHub repository (all notebooks + this presentation):
<https://github.com/AbhinavJag/spacex-falcon9-capstone>

Thank you!

